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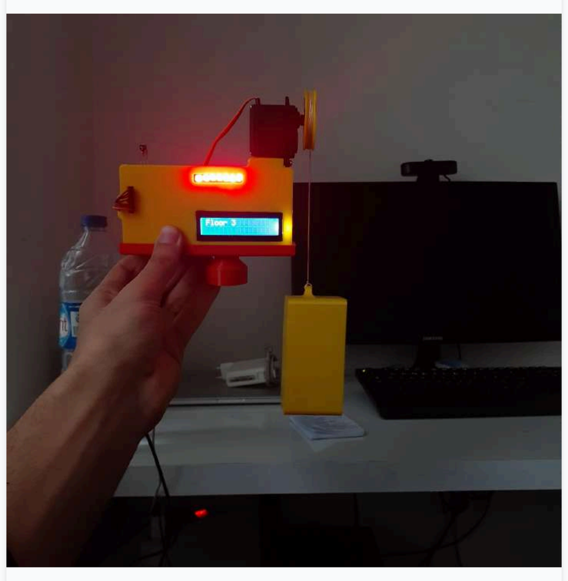
Projects

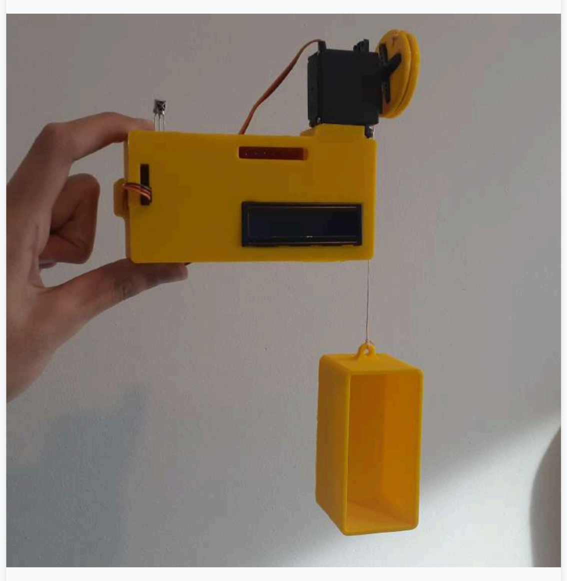
Air Defence competition at Teknofest



This project focused on designing and implementing an air defense system capable of autonomously and manually popping balloons from a distance of 6 meters. The system featured a PC-based user interface that enabled manual control of the platform and incorporated computer vision for autonomous targeting. The interface communicated with an STM32 microcontroller, which managed stepper motor control, body movement, and the laser firing mechanism. I was responsible for designing and assembling the complete electronic system, developing the motor control circuits, and integrating an absolute magnetic encoder for precise feedback. Communication between components was handled via UART and I²C protocols, while motor actuation was controlled using hardware timers to ensure accurate and responsive operation.

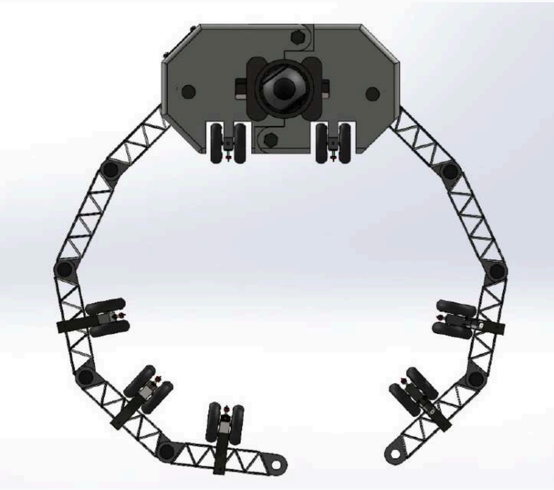
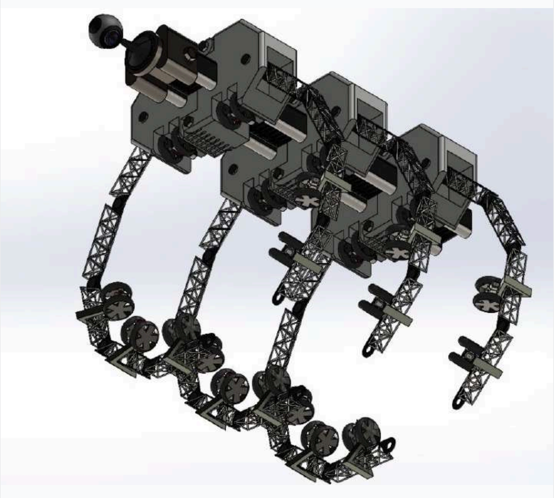
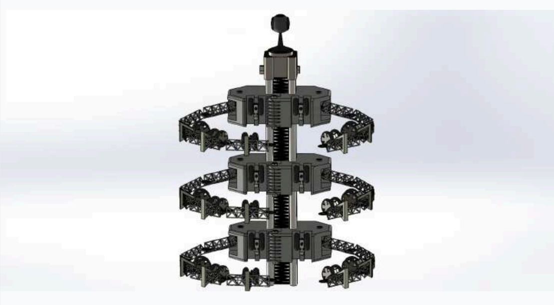
3-Floor Elevator System





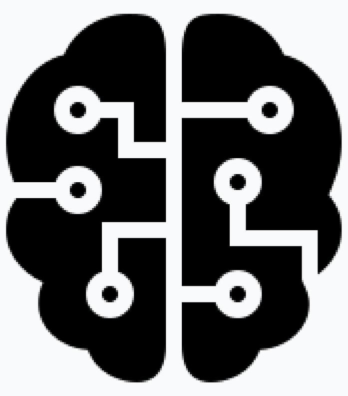
Designed and implemented a 3-floor elevator system with embedded control and IR remote operation, ensuring safe and efficient movement between floors. The system was built around a PIC16F887 microcontroller and a 360° servo motor capable of multiple rotations with precise position feedback. Floor selection and movement were controlled using IR sensors and a remote, providing intuitive and reliable operation.

SolidWorks 3D Model - Pipeline Inspection Robot

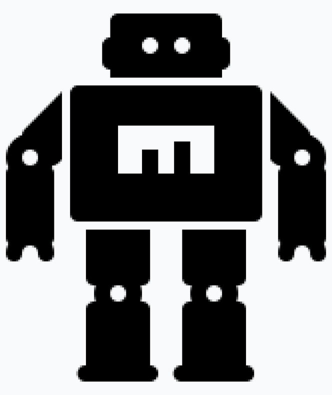
Developed a pipeline inspection robot designed to navigate along pipes while providing real-time visual inspection. The robot is equipped with a forward-facing camera for monitoring and analysis, and it is securely held in place by six adjustable arms that grip the pipe, ensuring stable movement. This project highlights mechanical design for secure attachment, mobility on cylindrical surfaces, and integration of embedded control with visual inspection capabilities.

Teknofest AI in Aviation



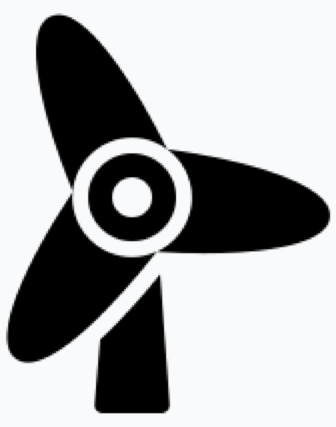
In this project, my teammate and I developed an AI-based system to analyze drone footage using a single camera. The first task involved object detection, where we trained a YOLO model to identify four object classes: humans, vehicles, and two types of flags. The second task focused on position estimation relative to the first frame, using only the drone footage. For this, we employed a combination of optical flow, SuperPoint feature extraction, SuperGlue matching, and essential matrix computation, with the Umeyama algorithm applied to determine scale. This project demonstrates the integration of computer vision techniques for object detection and precise position estimation using monocular drone footage.

Line Following Robot



This project involved designing and building a small line-following robot as an introductory robotics exercise. Using an Arduino microcontroller and multiple IR sensors, we developed a system capable of detecting and following a predefined path autonomously. The project provided hands-on experience in sensor integration, basic motor control, and embedded system programming, serving as a foundation for more advanced robotics projects.

Smart Wind Turbine



In this project, we developed a smart wind turbine system using an Arduino microcontroller. To reduce costs, we replaced traditional wind sensors with a lightweight object that acted as a mechanical wind vane, rotating freely with the wind. This object was connected to a potentiometer, allowing us to measure the wind direction by reading its angular position. Using this information, a motor automatically adjusted the turbine to align with the wind for maximum efficiency. This approach demonstrates creative problem-solving, cost-effective sensor design, and precise motor control in an embedded system.